

Kernel Assessment in High Dimensional Bayesian Optimization

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- $\{f(\mathbf{x})\}_{\mathbf{x} \in \mathbb{R}^d}$ is a **Gaussian process (GP)** if for all $\mathbf{x}_1, \dots, \mathbf{x}_n \in \mathbb{R}^d$, the vector $[f(\mathbf{x}_1), f(\mathbf{x}_2), \dots, f(\mathbf{x}_n)]^\top$ has a **multivariate normal distribution**.

$$f(\mathbf{x}) \sim \mathcal{GP}(\mu(x), k(\mathbf{x}_i, \mathbf{x}_j))$$

- **GP Regression:** $y(\mathbf{x}) = f(\mathbf{x}) + \epsilon_{noise}(\mathbf{x})$
- GP uses *MLE* to learn kernel hyperparameters (e.g., lengthscale ℓ , outputscale η^2)

Types of Kernels

- Matérn Kernel:

$$k_{\text{Matern}}(x, x') = \eta^2 \frac{2^{1-\nu}}{\Gamma(\nu)} \left(\sqrt{2\nu} \right)^\nu K_\nu \left(\sqrt{2\nu} d \right)$$

where $d = (x - x')^\top \Theta^{-2} (x - x')$, $\Theta = \text{diag}(\ell)$, and K_ν is the modified Bessel Function

- Squared Exponential (SE) Kernel:

$$k_{\text{RBF}}(x, x') = \eta^2 \exp \left(-\frac{1}{2} (x - x')^\top \Theta^{-2} (x - x') \right)$$

- **Bayesian Optimization (BO)** is a strategy to predict global optimum of black-box functions that are expensive to evaluate.
- 2 Parts: Gaussian Process and an Acquisition function
- Ex: Upper Confidence Bound:

$$\alpha(x; \lambda) = \mu(x) + \lambda\sigma(x)$$

- Explicit balance between **Exploitation** and **Exploration**

Algorithm 1 Outline of Standard Bayesian Optimization Algorithm

Place a Gaussian process prior on f by choosing $\mu(\mathbf{x})$ and $k(\mathbf{x}, \mathbf{x}')$

Evaluate f at n_0 points on some parameter space of potential points H .

Initialize training set $\mathcal{D}$ with these n_0 points and set $n = n_0$

while $n \leq N$ **do**

 Update the GP to be $f(\cdot)|\mathcal{D}$, the posterior conditioned on $\mathcal{D}$

 Let $x_n \in H$ be a maximizer of the acquisition function based on $f(\cdot)|\mathcal{D}$

 Evaluate $y_n = f(x_n)$

 Append (y_n, x_n) to $\mathcal{D}$

 Increment n

end while

return the best point according to the optimization objective

Figure: Standard Bayesian Optimization (SBO) Algorithm

1-D Example

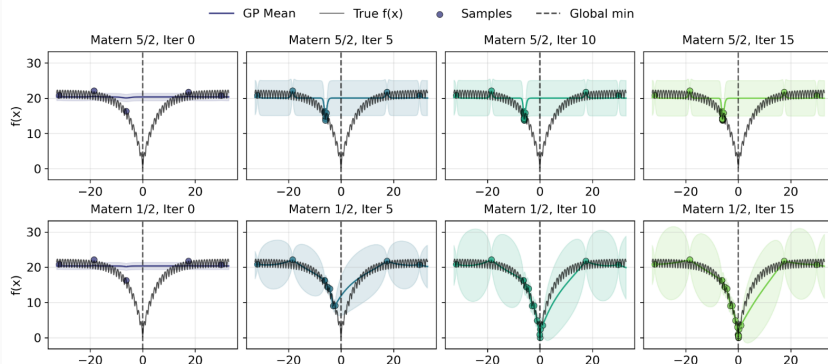


Figure: SBO on 1-D Ackley function with global minimum at $x = 0$

- SBO struggles in high-dimensions ($D > 20$) (Why?)
- **Curse of Dimensionality:** points become more sparse as input dimension increases $\rightarrow$ more difficult to model covariance structures
- Development of numerous High-dimensional methods but are expensive and complex

- Xu et al. (2025) pins issue to **gradient vanishing** [1]
- Ex: Squared Exponential Kernel

$$\left[\frac{\partial \mathbf{K}}{\partial \ell_k} \right]_{ij} = \frac{\partial \kappa_{\text{SE}}(\mathbf{x}_i, \mathbf{x}_j)}{\partial \ell_k} = \frac{2a}{\ell_k} \exp \left(-\frac{\|\mathbf{x}_i - \mathbf{x}_j\|^2}{\ell_k^2} \right) \cdot \frac{(x_{ik} - x_{jk})^2}{\ell_k^2} \leq \frac{2a}{\ell_k} \cdot \frac{\rho^2}{e\rho^2}$$

where x_{ik} and x_{jk} are the k -th element in $\mathbf{x}_i$ and $\mathbf{x}_j$ respectively

- Given any $\xi > 0$, $\frac{\rho^2}{e\rho^2} < \xi$ when $\rho > \tau_{\text{SE}} = \frac{1}{2} + \sqrt{\frac{1}{4} - \log \xi}$.
where $\tau_{\text{SE}} = 6.58$ when using float64
- With $\ell_0 = 0.5$ and $\ell_0 = 1$, the probability $p(\rho > \tau_{\text{SE}})$ exceeds 0.99 when $d \geq 205$ and $d \geq 473$ respectively

- Xu et al. (2025) remedies with robust initialization (RI) of the lengthscale proportional to dimension: $\ell \propto \sqrt{D}$
- Results show:
 1. Competitive or better performance to other novel HDBO methods
 2. Matérn 5/2 performs better and is more robust to gradient vanishing than SE

1. **No structural assumptions** on lengthscale
2. **Optimizer:** Xu et al. (2025) uses L-BFGS, we use Adam
3. Experiment across 6 benchmarks from Xu et al. (2025)

Additional Kernels Tested:

- Matérn 1/2
- General Cauchy Kernel
- Rational Quadratic

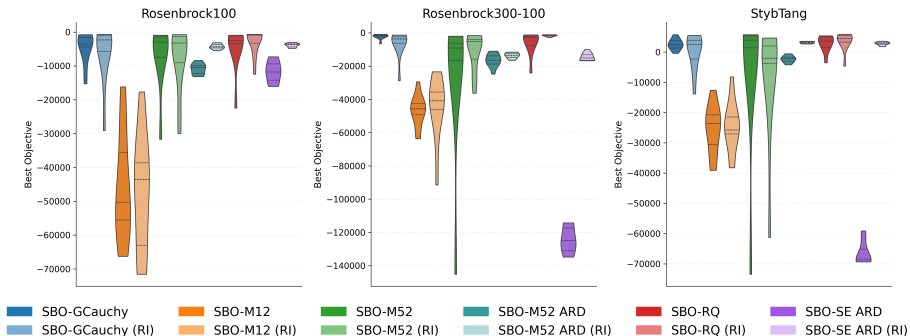


Figure: Final Value Across Synthetic Functions

Kernel Performance (Real-World)

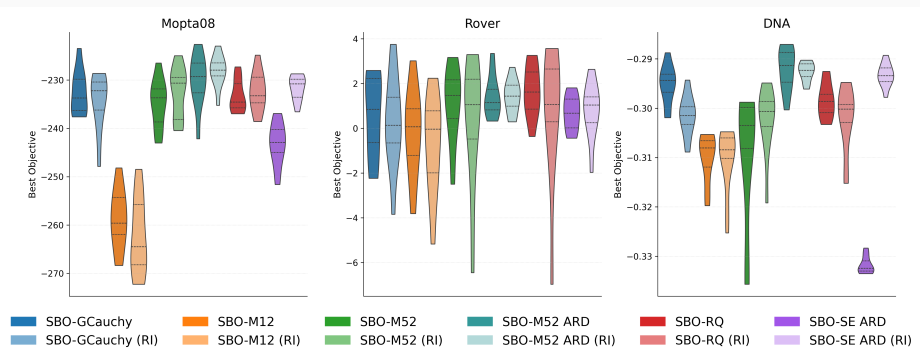


Figure: Final Value Across Real-World Optimization Tasks [2] [3] [4]

- General Cauchy Kernel:

$$k(x, x') = \left(1 + \left(\frac{\|x - x'\|}{\ell}\right)^\alpha\right)^{-\beta/\alpha}, \quad \alpha \in (0, 2], \quad \beta, \ell > 0$$

- Gradient w.r.t ℓ

$$\frac{\partial k}{\partial \ell} = \frac{\beta}{\ell} \cdot \rho^\alpha \cdot (1 + \rho^\alpha)^{-\frac{\beta}{\alpha}-1}, \quad \rho = \frac{\|x - x'\|}{\ell}$$

- Asymptotically:

$$\frac{\partial k}{\partial \ell} \sim \ell^{-1} \rho^{-\beta}$$

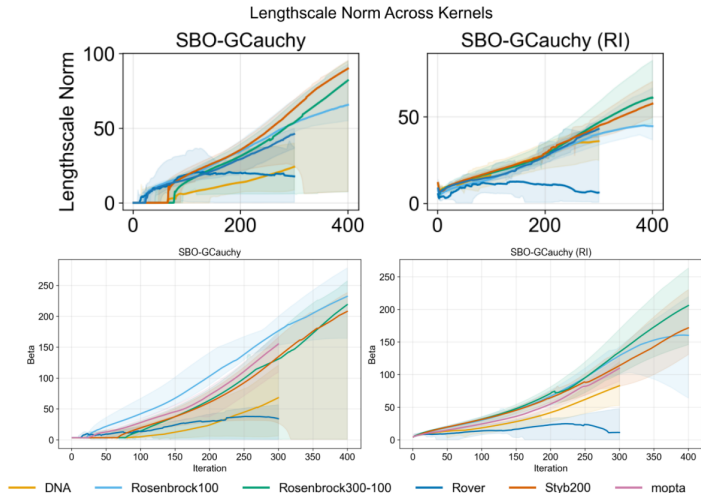


Figure: Hyperparameter Analysis for GCauchy Kernel

- [1] Z. Xu, H. Wang, J. M. Phillips, and S. Zhe, **“Standard gaussian process is all you need for high-dimensional bayesian optimization,”** in *The Thirteenth International Conference on Learning Representations*, 2025. [Online]. Available: <https://openreview.net/forum?id=kX8h23UG6v>.
- [2] D. R. Jones, **“Large-scale multi-disciplinary mass optimization in the auto industry,”** in *MOPTA 2008 Conference (20 August 2008)*, vol. 64, 2008.
- [3] Z. Wang, C. Gehring, P. Kohli, and S. Jegelka, **“Batched large-scale bayesian optimization in high-dimensional spaces,”** in *International Conference on Artificial Intelligence and Statistics*, PMLR, 2018, pp. 745–754.
- [4] K. Šehić, A. Gramfort, J. Salmon, and L. Nardi, **“Lassobench: A high-dimensional hyperparameter optimization benchmark suite for lasso,”** in *International Conference on Automated Machine Learning*, PMLR, 2022, pp. 2–1.

Thank You!

Questions?